accidents or violations remains a pressing issue, particularly as autonomous systems increasingly make independent decisions.

Addressing these challenges requires international cooperation, adaptive policymaking, and technological innovation. Initiatives such as digital airspace mapping, remote identification systems, and AI-assisted traffic management are paving the way toward smarter regulatory ecosystems. As drones evolve into integral components of smart cities, logistics networks, and defense systems, the future of drone regulation will depend on establishing a unified, flexible, and forward-looking governance framework that ensures both innovation and responsibility in global airspace management.

7.4.1 Navigating global regulatory frameworks for UAVs

The exponential growth of unmanned aerial vehicles (UAVs) has created an urgent need for robust and harmonized global regulatory frameworks that govern their safe and ethical use. However, the rapid pace of drone innovation has outstripped the ability of international and national authorities to create uniform standards. As a result, UAV operations today are regulated under a complex mosaic of national policies, aviation laws, and international guidelines, leading to inconsistencies that hinder global interoperability, research collaboration, and commercial scalability.

Most nations have established civil aviation authorities such as the Federal Aviation Administration (FAA) in the United States, the European Union Aviation Safety Agency (EASA) in Europe, and the Directorate General of Civil Aviation (DGCA) in India to oversee drone operations within their airspace. These agencies define rules for drone registration, pilot licensing, maximum flight altitudes, line-of-sight requirements, and operational restrictions near airports or sensitive zones. Despite these national efforts, there remains no universal legal framework that